

RESOLVE

An Agentic Architecture for Dispute Resolution in Regulated Consumer Finance

Bharathwaj Nandagopal

bharathwajna@gmail.com · MS CS (AI), Georgia Institute of Technology

August 2026

Independent research project built entirely on public data from the CFPB Consumer Complaint Database and public company filings.

Not affiliated with, or endorsed by, Capital One Financial Corporation. Live, self-refreshing demonstration and full source code accompany this paper.

Abstract

Consumer complaints against Capital One filed with the CFPB nearly tripled between 2021 and 2025 (12,519 to 34,102 per year), while 99.9% of them were answered “timely” — and 83% were closed with an explanation only. This paradox, visible entirely in public data, indicates that dispute handling fails on resolution quality rather than speed. We analyze 55,402 Capital One complaints (21,192 with consumer narratives) and find that 29% form a dispute/investigation-failure cluster in which one in eight customers describes re-submitting evidence they had already provided. We then present Resolve, a three-agent architecture for the dispute lifecycle — an unsupervised early-warning engine (Sentinel), a confidence-routed regulatory-lane triage classifier, and an LLM investigator copilot whose outputs pass through deterministic validation gates. A working prototype on the public corpus achieves 86% triage accuracy at 64% auto-route coverage, independently rediscovers a known regulatory incident from raw text, flags an emerging surge in evidence-ignored merchant disputes, and demonstrates citation-grounded letter drafting in which no claim can reference evidence that does not exist. We conclude with an evaluation framework, a value-pool sizing of \$85–100M/yr, and a governance-first rollout design intended for model-risk review in a regulated institution.

1 · Motivation: fast answers, unresolved problems

Capital One is among the most AI-forward banks in the United States, operating production multi-agent systems and deep ML in fraud and credit decisioning. Yet its complaint record has worsened faster than its business has grown: CFPB complaint volume rose from 12,519 (2021) to 34,102 (2025), with 21,300 filed through mid-August 2026 — placing it in the same complaint tier as retail banks two to three times its branch footprint. Purchase volume grew roughly 40% in the same window (partly via the Discover acquisition); complaint intensity outpaced it.

The telling statistic is not volume but disposition. Of 55,402 complaints since January 2025, 99.9% received a “timely” response — the metric a servicing organization dashboards — while 83% closed with explanation only and 4% with monetary relief. Speed is solved; resolution is not. Because internal metrics reward the former and rarely measure the latter, the failure mode is structurally invisible from inside — and fully visible in public data.

2 · Evidence from 55,402 public complaints

2.1 Data and method

We use the CFPB Consumer Complaint Database public API, filtered to Capital One Financial Corporation, January 2025 through August 2026: 55,402 complaints, of which 21,192 carry published consumer narratives (~38%; narratives require consumer consent and are scrubbed by the CFPB). Trend analyses use share-of-monthly-volume rather than raw counts to remain robust to publication lag, and exclude the two most recent calendar months.

2.2 Findings

Finding	Measure
Complaint growth 2021→2025	12,519 → 34,102/yr (2.7×); 2026 pacing ~34k
Timely response rate	99.9%
Closed with explanation only (since 2025)	83% · monetary relief: 4%
Dispute/investigation-failure cluster	29% of narratives

Finding	Measure
Customers describing re-submitted evidence (within cluster)	12.8%
Top issues since 2025	Incorrect info on report (10,934) · Managing an account (7,958) · Improper use of report (5,112) · Unresolved purchase disputes (4,850) · Failed investigations (3,819)

Narratives make the mechanism concrete. A traveler charged \$110 by an ATM that dispensed no cash: “Capital One has refused to issue a refund or investigate until the charge posts... no provisional credit or recourse while I’m abroad.” An identity-theft victim: “I submitted a dispute stating I never opened this account; they closed my dispute and stated the account came back verified.” A customer whose autopay linkage silently failed three times, generating derogatory credit reporting each time — a systemic defect surfacing as individual FCRA disputes. The common thread is not rude service but lost context: evidence and intent die at each handoff between intake, investigation, and furnishing.

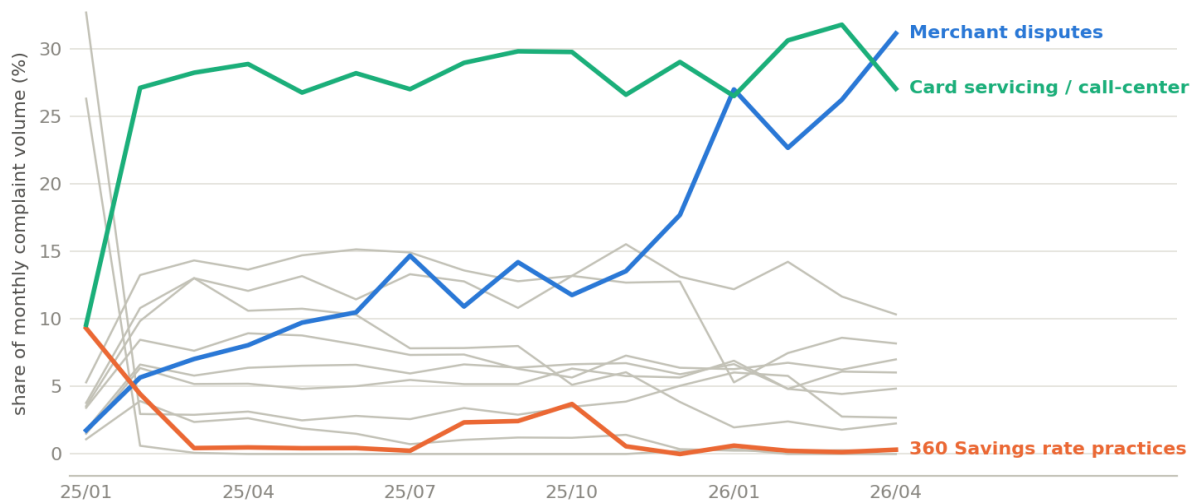


Figure 1 — Root-cause theme share of monthly complaint volume (12 unsupervised NMF themes; three highlighted, nine in gray).
“Merchant disputes — evidence ignored” more than doubles its share in the most recent observed quarter.

3 · The Resolve architecture

Resolve places three cooperating agents over the existing dispute lifecycle. It deliberately adds a layer over current systems rather than replacing them: the copilot is additive and never load-bearing, so any component failure degrades to today's workflow.

Agent	Function	Prototype implementation
Sentinel (risk-facing)	Clusters complaint/dispute streams into root-cause themes; flags emerging systemic issues before they become regulatory volume	TF-IDF + NMF (k=12); share-of-volume spike detection vs. baseline
Triage (intake-facing)	Routes each case to its regulatory lane (Reg Z, Reg E, FCRA, FDCPA, servicing) with calibrated confidence	TF-IDF + logistic regression; threshold-gated auto-routing
Investigator Copilot (associate-facing)	Assembles the case file, extracts an evidence inventory, drafts a citation-grounded response letter for human approval	LLM structured extraction + deterministic grounding check; constrained generation + citation validator

Sequencing follows risk and leverage: the associate-facing copilot ships first (a human signs every output, so no new customer-facing regulatory surface), the intake agent second (trained on the evidence taxonomy the copilot discovers), Sentinel throughout (it consumes the case exhaust the others produce). This ordering is a governance strategy as much as a product roadmap.

4 · Prototype results

4.1 Sentinel

With no labels, the topic model independently rediscovered the 360 Savings rate-practices controversy — subject of a \$425M settlement in 2025 — as a distinct theme, demonstrating that the method surfaces known regulatory incidents from raw text. It also detected an active anomaly: “merchant disputes — evidence ignored” grew from ~11% to ~26% of monthly complaint share (2.5–3.5 σ above baseline depending on window), precisely the failure mode Section 2 identifies. In production, such a detection would open a case with the payments-disputes team with member narratives attached.

4.2 Triage

Operating point	Coverage	Accuracy
All predictions	100%	75%
Confidence ≥ 0.5	88%	79%
Confidence ≥ 0.7 (auto-route)	64%	86%
Confidence ≥ 0.8	50%	89%

Macro-F1 is 0.73 across five lanes (held-out n=4,239). Accuracy rising monotonically with confidence is the property that makes threshold-gated routing safe: high-confidence cases auto-route; the remainder joins a human triage queue. The dominant confusion (FCRA $\leftrightarrow$ FDCPA) is domain-real — debt-collection complaints frequently allege credit-reporting harm — and argues for dual-lane tagging rather than a hard boundary.

4.3 Investigator Copilot and the validation gates

The copilot pipeline is: narrative $\rightarrow$ structured evidence extraction $\rightarrow$ drafted Reg Z acknowledgment letter $\rightarrow$ deterministic citation validation. Two design rules carry the safety argument. First, extraction returns each

evidence item with a verbatim quote, and plain code verifies the quote exists in the source text — an ungrounded item is dropped regardless of how plausible it sounds, which also neutralizes prompt-injection attempts embedded in narratives. Second, the letter drafter may cite only inventory items by tag; a validator resolves every citation and blocks any draft containing one that fails, feeding the failure back for a bounded retry before falling back to a blank template. In testing, deliberately corrupted drafts are blocked deterministically. The principle: gates are code, not prompts — zero-tolerance requirements receive engineering guarantees, not improved model averages.

5 • Evaluation framework

AI products fail evaluations in two directions: measuring the model while ignoring the business, or measuring the business with no way to attribute change to the model. Resolve specifies an eval ladder in which each rung must predict the one above it.

Rung	Measured	Method
Component	Citation resolution (gate: 100%); extraction accuracy	Golden set of adjudicated cases; every change
Task	Letter quality vs. human baseline	Blind QA rubric scoring, AI and human drafts interleaved
Product	Handle time; adoption; rejection reason codes	Pilot A/B (10 vs. 10 investigators)
Business	First-pass correct resolution; loss rate	Blind re-adjudication of matched grant and denial samples from both arms
Strategic	CFPB complaints per million accounts	Quarterly, lagging 6–12 months

Every optimization metric ships with a paired guardrail and a pre-written breach protocol. The canonical case: if copilot-assisted investigators grant monetary relief more often, the delta is either loss-rate creep or the correction of historical false denials — and only blind re-adjudication of both grants and denials, in both arms, can distinguish them. Deciding that protocol before launch removes the post-hoc interpretation fight. A pilot ships with a written kill criterion (e.g., <10% handle-time reduction $\Rightarrow$ do not scale) co-signed by the operations sponsor.

6 • Business case

Anchored to Capital One’s own Q4 2025 disclosures (quarterly purchase volume \$238.7B — roughly \$950B annualized post-Discover; quarterly non-interest expense \$9.3B; \$1.1B of Discover integration spend in 2025), a bottom-up model estimates ~10–19M annual dispute/fraud cases (0.05–0.1% of ~19B transactions) at a blended \$15–35 fully-loaded cost per case — a \$150–350M annual servicing cost base, plausibly 0.5–1% of operating expense.

Value pool	Mechanism	Annual value (base)
Servicing efficiency	25–35% handle-time reduction on the cost base	\$40–120M (base $\approx$ \$60M; pilot-provable)
Recovery uplift	Better-evidenced representment; fewer auto-conceded small-dollar write-offs	\$10–30M
Retention	Dispute experience is a leading switching trigger for fraud victims	\$15–40M (directional)

Value pool	Mechanism	Annual value (base)
Regulatory & complaint cost	~34k CFPB complaints/yr handling + tail-risk reduction (2025: \$425M settlement)	\$10M+ plus tail risk
Discover surge absorption	Capacity without hiring as two dispute stacks converge	Timing argument

A single product pod (~\$4–5M/yr fully loaded, including inference) breaks even on ~8% of the provable efficiency pool. The model's weakest assumption — the handle-time reduction — is flagged as low-confidence by design and is exactly what a one-quarter pilot exists to measure.

7 · Deployment as a living system

The accompanying demonstration is not a static snapshot. A scheduled GitHub Actions workflow re-downloads the latest Capital One complaints from the CFPB public API weekly, re-runs the full pipeline (classifier retraining, theme clustering, spike detection, copilot case generation), rebuilds the dashboard, and republishes it — so Sentinel's spike alerts track the real, current complaint stream with no servers and no exposed credentials. (The CFPB API rejects browser-origin requests, so refresh runs server-side on scheduled runners; the optional LLM step reads its key from encrypted repository secrets.) The same loop is the production pattern in miniature: scheduled retraining, versioned outputs, and a dashboard that is always as fresh as its source.

8 · Limitations and future work

CFPB complaints are the visible tip of internal dispute volume (plausibly 0.2–0.3% of cases), and published narratives are a consented, scrubbed ~38% subsample; both may bias theme composition. Regulatory-lane labels derive from the CFPB issue taxonomy rather than adjudicated ground truth. The public demonstration precomputes LLM outputs rather than serving live inference (a deliberate key-safety choice); the letter templates model one lane (Reg Z acknowledgment) of many. The sizing model's case-volume and cost-per-case assumptions are industry heuristics awaiting internal validation — the assumptions ledger marks each with its confidence and validation path. Future work: Reg E and FCRA letter lanes, an in-browser triage demonstration, retrieval over policy documents for citation-grounded regulatory references, and a human-scored extraction eval against the pattern baseline.

References

1. CFPB Consumer Complaint Database, public API. consumerfinance.gov/data-research/consumer-complaints/ (accessed Aug 2026).
2. Capital One Financial Corporation, Fourth Quarter 2025 Results, January 22, 2026. investor.capitalone.com.
3. CFPB, Consumer Response Annual Report 2024 (May 2025).
4. VentureBeat, "Why Capital One built its multi-agent AI platform around open-weight models" (2026).
5. Resolve source code, test runbook, and living demonstration. github.com/bj-nanda/resolve (live demo: bj-nanda.github.io/resolve).